Lecture 18

Tuesday, 21 March 2023 1:29 PM

Silent Spring, Rachel Carson → "lucid" communication

Current Distributions

Reaction Rate is a key design parameter for any chemical rxn eng. analysis and we know for non-echem rxns, it depends on c_j, T

$$r_j = f(c_j, T, P) \text{ for non-echem}$$

In e-chem systems, $r = \frac{i}{nF}$, and we know that i is additionally dependent on V & ϕ

$$i = f(c_j, T, P, V, \phi)$$

∴ This can create spatial differences of rxn rate that we might not predict from classic rxn eng. (have to solve PDE's for each finite element)

↳ Need some simplifications that arrive at approx. for real behaviour.

+ understand scenarios that lead to specific regimes

+ " governing eqs + BC's within each regime

+ Generalities about current dist., based on solutions to gov. eqs.

Two dim. parameters:

① $\frac{i_{avg}}{i_{lim}}$ → spatial avg (@ each position) ⇒ describes imp. of conc. gradients

↳ $\frac{i_{avg}}{i_{lim}} \to 1$ (MT-controlled limit)

↳ ∇c is important

↳ $\frac{i_{avg}}{i_{lim}} \to 0$ (uniform composition)

↳ ∇c not important

② $W_a = \frac{R_{ct}}{R_e}$ ⇒ describes imp. of charge transfer vs. Ohmic resistance

↳ different dependence on i, depending on kinetic expression (tafel, or linear)

General case:

Governing Eqs: $N_j = -z_j u_j F c_j \nabla\phi - D_j \nabla c_j + v c_j$

$\sum z_j c_j = 0$

→ Solving this gives us the complete picture

"Tertiary Current distribution"

1) This occurs when $\nabla c_j \neq 0$ ($i_{avg} \not\approx i_{lim}$)
2) Rct an Re are both imp
3) $t_j \to 0$ ⇒ convection diffusion (fast kinetics) → i_{lim}
4) $t_j \neq 0$ ⇒ full governing eqs (no simplifications)
5) $W_a > 0.1$
6) MT BC's typically

Can ∇c_j be neglected? ($\frac{i_{avg}}{i_{lim}} = ?$)

No → $\frac{i_{avg}}{i_{lim}} \sim 1$

Yes ($\frac{i_{avg}}{i_{lim}} \to 0$) → Governing Eq: (Laplace) $\nabla^2\phi = 0$

Can electric field be neglected? ($t_j = ?$)

No ($t_j > 0$)

Yes, $t_j \to 0$ → Gov. eqs: Convection diff. eq.

$$\frac{\partial c_j}{\partial t} + \bar{v}\cdot\nabla c_j = D_j \nabla^2 c_j$$

Should Rct be considered? ($W_a = ?$)

No ($W_a \to 0$) → Primary Current Distribution

1) Occurs when $\nabla c_j = 0$ (uniform conc)
2) Ohmic resistance dominates

↳ Result in constant potential BC's → linear BC's + governing eqs

Yes ($W_a > 0.1$) → Secondary Current distribution

1) $\nabla c_j = 0$
2) Ohmic + Ct resistances are both imp. ($W_a > 0.1$)

↳ (i-η) type BC [tafel, linear etc] (kinetic BC)

↳ Always more uniform than primary

Further considerations: Peclet No.

↳ Result in constant potential BC's → linear BC's + governing eqs & non-uniform current dist. (has the maximum non-uniformity)

Further considerations regarding convective transport: Peclet No.

Regions:

Polarization curve → There is i_{lim} for anodic & for cathodic (these two are different)

i_{lim}

III → high kinetics (chemically dominated) R_{ct} is small [Primary]

II → Secondary?

I

Don't have enough info about R_Ω to make the general claim

→ large WE

→ electrolyte solution has poor conductivity

$K \downarrow$ $R_\Omega \uparrow$, } moving towards primary

Large electrodes + high conc'n

↳ significant non-uniformity